INCH-POUND

MIL-P-70525 (AR)
AMENDMENT 6
29 August 2002
SUPERSEDING
AMENDMENT 5
22 July 1996

MILITARY SPECIFICATION

PROJECTILE, 105MM, HE, RA, XM913/XM927
MOTOR BODY BONDING ASSEMBLY FOR

MIL-P-70525 (AR) was inactivated after 21 August 1998 for new design.

This amendment forms a part of MIL-P-70525 (AR), dated 29 October 1990, and is approved for use by the U.S. Army Armament, Research, Development, and Engineering Center, and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 1

Title:

Change "XM913/XM927" to "M913/M927", wherever it appears.

Change "Projectile" to "Cartridge"

1.1 Line 4, Change "(See 6.11)" to "(See 6.1)"

PAGE 2

* 2.1.1, Delete "MIL-P-116", "MIL-P-117", "MIL-R-13562" in their entirety.

3.2, Delete "939 1136" and substitute "938 1136"

3.3.2, Capitalize "there"

3.3.3, Change "Cracks" in two places and substitute "Discontinuities"

3.3.5.1, Delete (Cracks) in two places and substitute "Discontinuities"

PAGE 5

3.3.5.3, Delete "MIL-R-12562" and substitute "MIL-R-13562"

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* 3.3.5.3, Delete in its entirety and substitute the following :

"3.3.5.3 Surface voids. Surface voids shall not exceed 1/16 inch in greatest dimension, the total number of voids in a square inch area shall be less than 10 with a total area of less than .25 square inches."

3.7, Delete in its entirety and substitute the following:

"3.7 Proving ground test. There shall be no motor body bonding assembly related premature explosive function, collapse, separation or break-up of metal parts in the gun bore or in flight. Testing shall be as specified in 4.5.8."

3.8.2, Delete "(See Figure 1)" and substitute "(See Figure 3)"

PAGE 10

Add new Figure 3 as included page 10a.

PAGE 12

4.3.1.1:

Line 3, change "4.32" to "4.3.1.2"

Change "Finished Machined Motor Bodies Unpainted and Unknurled" to "Mechanical Properties"

Delete, "4.4.3.2/4.5.5" and substitute "4.4.3.1/4.5.7"

PAGE 14

Table I:

Under "Finished Machined Motor Bodies Unpainted and Unknurled", Delete"4.4.3.2/4.5.5" and substitute "4.4.3.1/4.5.7"

Under "Nozzle Inserts, Compression Test", Delete "3.6.1" and substitute "3.8.1"

Under "Radiographic Inspection", Delete "3.6.2" and substitute "3.8.2"

PAGE 17

4.4.2.1, Delete the title in its entirety and substitute "Motor Body, Prior to Weld"

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X-RAY INSPECTION ZONES

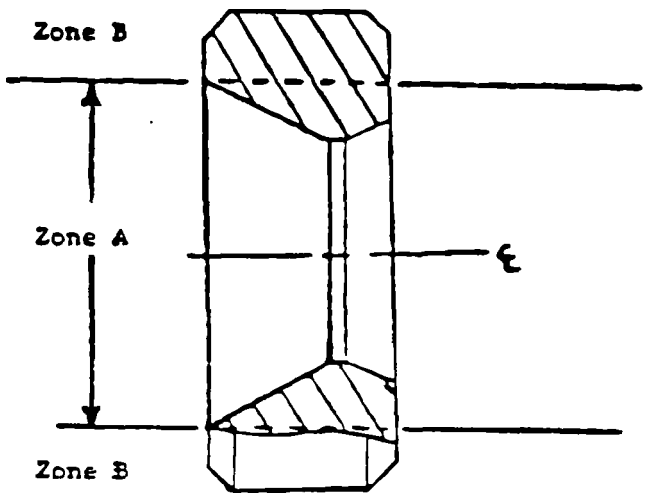


Figure 3. X-ray of nozzle.

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4.4.2.2 :

Delete title in its entirety and substitute “Motor Body, Protective Coating”

Add the following Specials:

“2. Ultrasonic C-Scan test	100%	3.6	4.5.4
3. Ultrasonic C-Scan test stamp missing	100%	3.2	Visual”

Major 107, Add “(if applicable)”

PAGE 20

4.4.2.2:

Delete paragraph number “4.4.2.2” and substitute “4.4.2.3”

Delete Major defects 102 and 103 in their entirety and substitute the following:

“102. Salt spray test	4.4.3.6.1	3.2	4.5.5.1
103. Protective coating and thickness	4.4.3.6.2	3.2	4.5.5.2”

Major 104, Change “3.10” to “3.2”

Major 134, Delete in its entirety.

Delete Major defects 142 and 143 in their entirety and substitute the following:

“142. Mechanical properties of body	4.4.3.1.2	3.2	4.5.7
143. Properties of rotating band weld	4.4.3.2	3.2	4.5.2”

Add new minor defect 211 as follows:

“2 11. Radius at cavity base	Level V	3.2	4.4.3.7/Gage”

PAGE 23

4.4.2.4:

Delete Major 101, 102, 103, and Minor 201 in their entirety and substitute the following:

“10 1. Radiographic inspection	100%	3.8.2	4.5.3.2
102. Grain flow	4.4.3.5.1	3.2	4.4.3.5.2

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103. Compression test	4.4.3.5.1	3.8.1	4.5.3.1
201. True position of throat diameter with outside diameter	Level V	3.2	Gage”

Major 105, Delete “(2 Places)”

Add Major 111 as follows:

“111. Angle, aft end	Level	III	3.2	Gage”

PAGE 24

4.4.2.5:

Delete Major defect 101 and 102 in their entirety and substitute the following:

“101. Adhesive not present on forward O.D. nozzle edge or rear nozzle I.D. edge (360 degrees coverage required)	100%		3.2	Visual
102. Gap between aft end of nozzle and motor body	Level	III	3.2	Gage”

4.4.3.2.1:

Delete the second and third sentences and substitute the following:

“The chips for the sample shall consist of the copper overlay between 4.20-.01 and 4.223-.01 inches diameter (full width). All of the chips shall be collected, crushed, and dissolved (see 6.8.1).”

Sixth sentence, delete “samplie” and substitute “sample”

PAGE 30

4.4.3.8.1:

Delete “4.5.9” and substitute “4.5.8”

Change “3.10” to “3.7”

4.4.3.8.2:

Add before the last sentence: “If the lot is rejected, the acceptance criteria reverts back to the beginning of the paragraph.”

Delete “4.5.9” and substitute “4.5.8”

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Change “3.10” to “3.7”

PAGE 33

4.5.4.1, Second paragraph, first sentence, delete “reviuosly” and substitute “previously”

PAGE 34

4.5.5.1, Delete “4.4.3.5.1” at the end of the paragraph and substitute “4.4.3.6.1”

4.5.8, Delete in its entirety and substitute the following:

“4.5.8 Proving ground. Ballistic testing shall be performed at a Government proving ground in accordance with the applicable acceptance test procedure. The sample motor body assemblies selected in accordance with 4.4.3.8 shall be live loaded with rocket propellant grain and delay assemblies, then assembled with TNT loaded warhead and conditioned to 145-5 degrees F for 16 hours minimum prior to firing rocket on in a Ml 19 series howitzer using a PXR200 propelling charge conditioned to 70 ± 2.5 degrees F. The warheads shall be fuzed with M577 series MTSQ fuzes set to 40 seconds selected by the proving ground. Photographic and radar coverage shall be provided to assure compliance with 3.7. Range and deflection shall be recorded for informational purposes only.”

(Note paragraph 4.5.8.1 added in amendment 4 has been deleted by amendment 5)

PAGE 35

4.5.9, Delete in its entirety.

5.1, Add the following:

“with MIL-STD-1169, Level C. Appropriate cushioning and partitions shall be provided to protect the items from contracting each other during shipment. Sufficient fillers must be used to insure a tight pack. Adequate corrosion protection shall be provided, if required.”

6, Notes, add the following parenthetical note:

“(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)”

6.2, Delete in its entirety and substitute the following:

“6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of the specification.
b. Issue of DODISS to be cited in the solicitation, and if required, the specific issues of individual documents referenced (see 2.1.1)”

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6.3, Delete in its entirety and substitute the following:

"6.3 Consideration of data requirements. The following data requirements should be considered when this specification is applied on a contract. The applicable Data Item Descriptions (DIDs) should be reviewed in conjunction with the specific acquisition to ensure that only essential data are requested/provided and that the DIDs are tailored to reflect the requirements, a Contract Data Requirements List (CDRL) (DD Form 1423) must be prepared to obtain the data, except where DOD FAR Supplement 27.475-l exempts the requirements for a DD Form 1423.

Reference paragraph	DID Number	DID Title	Suggested Tailoring
4.4.4	OT-90- 12009	AIE Design Documentation	None

The above DIDs were those cleared as of the date of this specification. The current issue of DOD 5010.12L, Acquisition Management systems and Data Requirements Control List (AMSDL) must be researched to ensure that only current, cleared DIDs are cited on the DD Form 1423."

6.4 Delete in its entirety and substitute the following:

"6.4 Submission of inspection equipment for design approvals. Gage designs as required shall be submitted to: M913/M927 QUAL. ENGR. (AMSTA-AR-QAA-R), US ARMY ARDEC, PICATINNY ARSENAL, NJ 07806-5000. This address shall be specified to the Contract Data Requirement List, DD Form 1423 in the contract."

6.7, Change "AMSMC-QAA-Q" to "AMSTA-AR-QAA-R" and "SMCAR-FSA-IM" to "AMSTA-AR-FSA-M"

6.8.1, Add the following new paragraph :

"6.8.1 Chemical analysis. An equivalent method may be submitted to the Government for approval."

The margins of this amendment are marked with an asterisk or vertical line to indicate where changes (additions, modifications, corrections, deletions) from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

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